# Robust Charging Infrastructure Design under Demand Uncertainty

*A solar-powered charging station for a shared e-micromobility fleet — Executive Summary*

Competition Themes — 3 (primary): solar-PV charging-station design · 2 (secondary): charge/discharge profiles under demand scenarios

| **20 kWp PV · 0 kWh · 8 chargers** | **-61% (£87,422/yr)** | **96% (vs 85% naive)** | **16% (4.6 tCO2/yr)** |
|---|---|---|---|
| Recommended robust design | Value of robustness (worst-case cost) | Guaranteed fleet service, worst case | Operational carbon saving |

## 1. Problem Understanding *[20% criterion]*

A shared-micromobility operator depot charges a mixed fleet of e-scooters, e-bikes and e-cargo bikes whose demand swings with season, weather, events and multi-year growth. This creates a sizing dilemma for a solar charging hub: one built for **average** demand fails at peaks and strands the fleet, while one built for the **worst case** sinks capital into rarely-used equipment. The dilemma sharpens under a **constrained grid connection** — a higher import limit needs a costly DNO reinforcement — so the evening charging peak must be served from on-site solar and storage, across enough charge bays. Sizing becomes a decision under deep uncertainty: under-build and lose service; over-build and waste money. Demand is anchored to the REAL DfT Newcastle e-scooter trial (monthly trips, fleet size, deployment, trip distance), so the problem is posed on observed data, not assumption.

## 2. Relevance & Innovativeness of the Proposed Solution *[20% criterion]*

The solution directly delivers **Theme 3** (design of a solar-PV charging station for a shared fleet) and **Theme 2** (modelling charge/discharge profiles under demand scenarios), producing an actionable station specification with uncertainty bounds. Its novelty is methodological: charging-station sizing is almost always **deterministic** (a single demand forecast). We instead apply **robust optimisation — minimax-regret, maximin and CVaR — alongside a two-stage stochastic program**, choosing a design that performs across the entire demand space, and we define and quantify the **"value of robustness"**: the worst-case cost and lost service avoided by hedging. A correlated Monte-Carlo "demand regime" reproduces the realistic co-movement of trips, utilisation and growth, and exposes a subtle but important result — expected-value design **under-weights the correlated high-demand tail**, which is precisely why scenario-based robust methods are required here. A further, counter-intuitive finding falls out: because depot charging is flexible, **smart charging is the cheapest robustness lever** — it flattens the load below the connection so that, at a grid-connected depot, a battery does not pay; storage becomes essential only as the connection weakens toward off-grid (~10 kW), a boundary we quantify. The recommendation is therefore solar + smart-managed bays, not capital-heavy storage.

## 3. Approach — Methodology, Algorithm & Flow *[20% criterion]*

The pipeline (Figure 1) is: **(i)** build Low/Medium/High demand scenarios from the real trial data, scaled across a year-on-year growth range into nine probability-weighted scenarios; **(ii)** simulate an 8,760-hour energy balance for any design under the capped grid (PV → battery → capped grid → unmet demand), with every reported design re-optimised under an LP-optimal rolling-horizon dispatch; **(iii)** evaluate all 150 designs (PV 5–25 kWp × battery 0–50 kWh × 4–20 chargers) against the nine scenarios, pricing unmet demand as lost service, to form a cost matrix; **(iv)** apply five decision rules — naive, a two-stage stochastic program (expected cost), CVaR (risk-averse), minimax-regret and maximin — and map the cost-vs-robustness Pareto frontier; **(v)** propagate uncertainty through a 500-sample correlated Monte-Carlo fan (with bootstrap 95% confidence intervals) and rank drivers with a tornado and global Sobol sensitivity; **(vi)** confirm the conclusion via a robustness-of-robustness sweep (it holds across every penalty x grid combination) and validate seven outputs against published benchmarks. **Output / GUI:** results are delivered as an interactive multi-tab dashboard — a Station Designer with live PV/battery/charger/grid sliders showing energy balance, service, cost and carbon in real time, plus Robust-Optimiser, Uncertainty and Sensitivity tabs — and a one-command HTML report. The full study runs in under a minute.

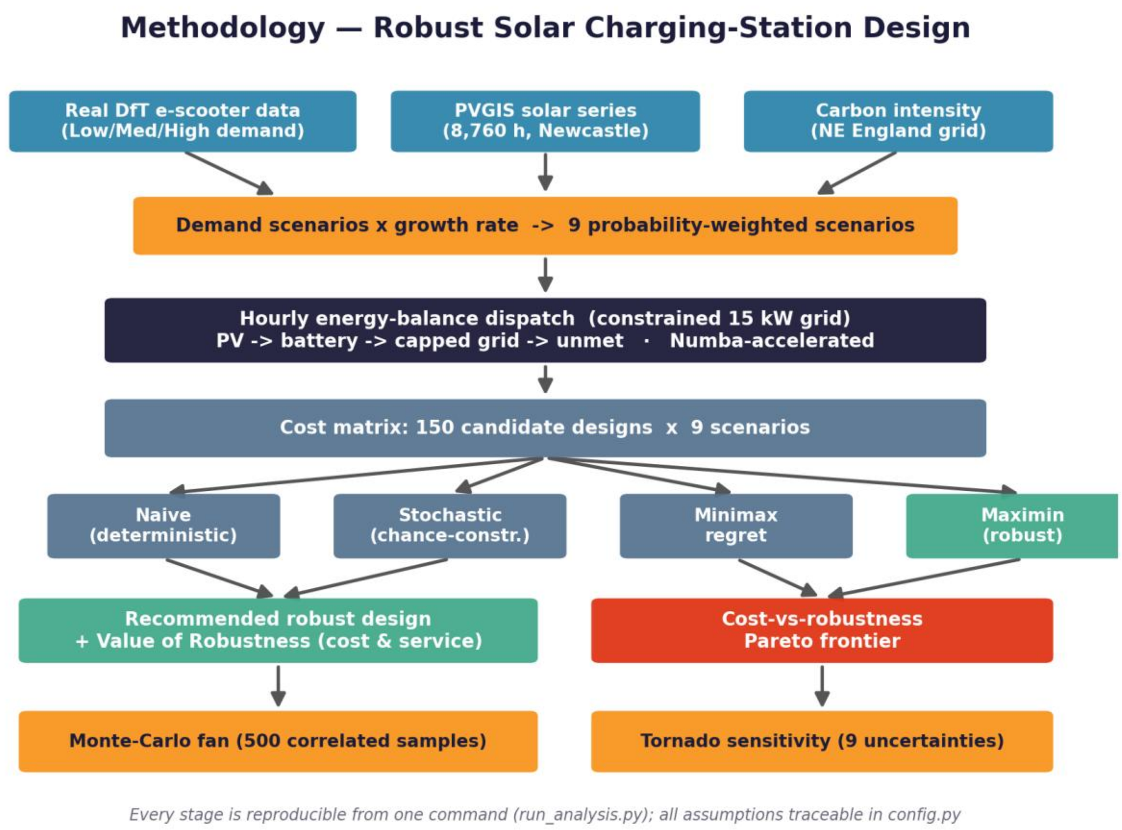


*Figure 1 — End-to-end methodology / algorithm flow.*

## 4. Feasibility of the Proposed Approach *[20% criterion]*

**Technical:** only commercially available components — monocrystalline PV, Li-ion storage and standard AC charge points, sized within datasheet-supported ranges. **Economic:** ≈£36,000 capital cost with bounded annual cost even in the worst demand scenario; the model reports CAPEX, OPEX, replacement and grid costs. **Operational:** one deployable specification plus the confidence interval around it. **Validated:** all seven key outputs fall within published benchmark ranges. **Battery sustainability & safety:** where the weak-grid case requires storage we specify a cobalt/nickel-free LFP pack (thermal-runaway onset ~270°C vs ~150°C for NMC, charged at a conservative 0.5C); the modelled duty cycle gives ~12-yr life, ~45% end-of-life material recovery and a second-life stationary phase before recycling. **Computational:** open-source Python, deterministic, one-command rebuild, passing test suite — any reviewer can reproduce every number.

## 5. Originality & Authenticity (AI / Plagiarism) *[20% criterion]*

All model code, the dispatch engine, the optimisation formulation and every figure were written from scratch for this submission. All three datasets are REAL: DfT e-scooter monitoring data (Open Government Licence), hourly solar pulled live from the EU PVGIS API, and regional grid-carbon intensity from the National Grid ESO API. Every numerical assumption carries an inline source citation and all literature is referenced. The repository is fully self-contained and reproducible — fixed random seed, one-command rebuild, passing tests — directly supporting the competition's AI and plagiarism checks. No third-party text or code is reproduced.


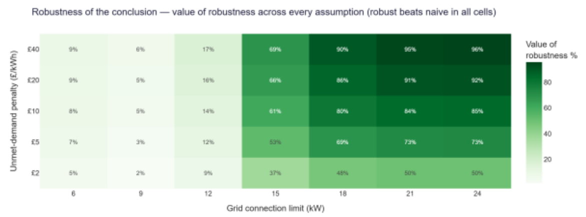


*Figure 2 — Cost-vs-robustness Pareto frontier (five decision rules). Figure 3 — Robustness of the conclusion: robust beats naive across every penalty x grid combination.*